



Invited review

CATastrophic myths part 1: Common misconceptions about the social behavior of domestic cats and implications for their health, welfare, and management

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ABSTRACT

Cats are among the most popular pets worldwide, but there are still major gaps in the public's general understanding of their social behaviors and related needs, including for socialization. In addition to these knowledge gaps, people often have negative or ambivalent attitudes about cats, which can directly impact their welfare outcomes. Insufficient attention to the behavioral ecology and development of cat sociality, along with failure to account for their highly variable individual preferences and tolerance for social behaviors can lead them to experience distress that undermines both their welfare and the human-animal bond. As Part 1 of a two-part series addressing common myths about cats, the purpose of this first paper is to review and debunk common misperceptions about the social needs and behaviors of cats, including misunderstandings about their social lives and abilities to bond with humans. We also identify where opportunities exist to improve socialization of cats and to advance research in related areas that might better support their behavior and welfare needs.

Introduction

Individual and cultural perceptions of the domestic cat (*Felis silvestris catus*) are diverse and have evolved over time. For example, cats were once revered as gods in ancient Egypt, but they were positioned as feared familiars of witches and the devil in medieval times (Serpell, 2000). Today, cats are increasingly viewed as friends, companions, and family members within homes around the world (Arañuri et al., 2017). As of 2021, 45.3 million households in the United States (US) reported keeping one or more companion cats, ranking them among the most popular companion animals in North America¹. Although a large and growing segment of the population in the US and worldwide thinks of cats as family members (Bir et al., 2016; Grigg and Kogan, 2019), cat behavior remains relatively poorly understood and the available scientific information is not always in the public domain where cat owners

and subsequently, cats can benefit from it.

In the absence of adequate scientific knowledge, attitudes towards cats, and interpretations of their behavior have often surfaced as more negative or ambivalent than those directed towards dogs (Kellert and Berry, 1980; Perrine and Osbourne, 1998; Bir et al., 2016). For instance, Kellert and Berry (1980) found that the U.S. public ranked cats as the 12th most liked domesticated species - much lower than horses and dogs. Perrine and Osbourne (1998) reported that 40% of people surveyed characterized themselves as 'dog people', while only 19% self-identified as preferring cats. However, there is also evidence that perceptions and preferences for cats may be shifting. For example, multiple studies conducted around the world suggest that cats are increasingly viewed as members of the family or children (Arañuri et al., 2017; Bouma et al., 2021) and are typically ranked as the second most popular pet². In a review reconsidering the archaeological history and

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¹ See: American Pet Products Association 2021–2022. https://www.americanpetproducts.org/press_industrytrends.asp (Accessed 29 August 2023).

² U.S. pet ownership statistics. (2020). American Veterinary Medical Association. <https://www.avma.org/resources-tools/reports-statistics/us-pet-ownership-statistics/> (Accessed 29 August 2023).

ancestry of the domestic cat, Driscoll et al. (2009) suggested that 'the aloof and elusive nature of cats' is a unique perception of the species that people may find both charming and infuriating. This notion of the cat as an independent entity, engaged in symbiotic, but not necessarily affectionate, relationships with humans is alluded to in Kipling's (1902) story, *'The Cat Who Walked by Himself'*³, where the cat is represented as 'the wildest of all wild animals' and describes himself as 'not a friend, and not a servant.' While diverse socio-cultural, geographical, relational, and personal factors may drive human views about cats, one constant is that cat welfare remains tied at least in part to human perceptions of them and their behavioral needs. Thus, assumptions and beliefs about cat behavior can have welfare consequences, even when inaccurate.

Animal welfare has been defined as '*... how an animal is coping with the conditions in which it lives. An animal is in a good state of welfare if (as indicated by scientific evidence) it is healthy, comfortable, well nourished, safe, able to express innate behavior, and it is not suffering from unpleasant states such as pain, fear, and distress*'⁴. Ensuring that cats are not only coping with their environments, but that their needs are met relative to the five domains of welfare (nutrition, environment, health, behavior, and mental states; Mellor and Beausoleil, 2015), requires understanding their ethology, evolutionary history, health, and ecology. Multiple studies suggest that many people who keep cats lack the basic knowledge to optimize their environments, the ways in which they are managed, and their overall well-being (Howell et al., 2016; Stella and Croney, 2016; Grigg and Kogan, 2019). In one study, a number of domestic cat owners appeared unable to interpret the behavior of their pets, particularly as it pertained to gauging their stress responses (Mariti et al., 2017). For example, 19% of owners failed to recognize house-soiling (urinating in an inappropriate place) as a sign of distress in their cats and 57% did not view aggression as indicative of distress. Lack of information and misinformation potentially jeopardize cat welfare. While comparing cat owners' and veterinarians' perceptions of cat behavior and welfare needs Bir et al., (2016) found that owners were inclined toward agreement that '*cats are independent*' and '*have less social needs than other animals*'. Beliefs such as these are not supported by the scientific data on cat social behavior and may result in failure to meet cats' individual social or environmental needs. Further evidence of public misunderstanding of cat behavior was provided by Grigg and Kogan's (2019) finding that owners who agreed with the sentiment that cats '*often misbehave to get back at their owners for doing something that the cat doesn't like*' were more likely to use positive punishment to deter unwanted behaviors. Unfortunately, positive punishment of cats (such as responding with loud verbal reprimands or hitting of cats in response to unwanted behaviors) is counterproductive, eliciting or exacerbating behavior problems such as biting or scratching (O'Hanley et al., 2021).

There is clear need to address common misconceptions about domestic cats and the implications for their behavior, health, welfare, and management. The purpose of this first paper is to review and debunk misunderstandings about the social needs and behaviors of cats, which can affect their welfare in homes, catteries, laboratories, and other environments (Crowell-Davis et al., 2004). We intend to encourage deliberation about the origins of widely held 'knowledge' about the domestic cat's sociality and ways in which robust, contemporary, science-based information can be created and widely disseminated to support cat welfare.

³ See: Kipling, R., The Project Gutenberg EBook of Just So Stories <https://www.gutenberg.org/files/32488/32488-h/32488-h.htm> (Accessed 29 August 2023).

⁴ See: World Organization for Animal Health. https://www.woah.org/fileadmin/Home/eng/Health_standards/tahc/2016/en_chapitre_aw_introduction.htm (Accessed 29 August 2023).

Myth 1: Cats are not social with each other

Over the last several decades, there has been increased research on the social behavior and cognition of dogs (Hare et al., 2002). The domestic dog's success at living in anthropogenic environments, and especially as pets in human homes, has often been attributed to their social cognition (Hare and Tomasello, 2005), gregariousness (vonHoldt et al., 2017) and the ease with which they form bonds with humans (Gácsi et al., 2001). Over this same period, the behavior and cognition of cats has received considerably less scientific attention (Vitale Shreve and Udell, 2015). Meanwhile, the number of cats living in homes as pets has continued to increase². In the US, there are now more domestic cats living in homes as pets than there are pet dogs, and multi-cat households are a significant driver of the trend for increased cat ownership². Fortunately, scientific interest in the social behavior of cats has recently increased, resulting in a strong and growing body of knowledge in this area (for an in-depth, cross-species comparative review of cat sociality, see Bradshaw, 2016). Critically, the data paint a very different picture of cat social awareness, behavior, and needs.

Domestic cats, like domestic dogs, are facultatively social, meaning that their social behavior is flexible and heavily influenced by their early development and lifetime experiences (Crowell-Davis et al., 2004; Bradshaw, 2016; Vitale et al., 2019). The majority of the world's cats are free-living and they may be solitary, live in large colonies with other cats, or fall somewhere in between (Crowell-Davis et al., 2004). The ability of cats to co-exist and cope well with each other in colonies appears to be dependent on the availability of resources, such as shelter, food (whether prey or provided by humans), and territory (Flockhart et al., 2016). Perhaps because cats have frequently been described as solitary (Kleiman and Eisenberg, 1973), there has been little investigation into their social behavior with other cats in the home environment (Bernstein and Strack, 1996). Many studies of domestic cat sociality have instead focused on semi-naturally occurring groupings of cats in outdoor colonies (Vitale, 2022).

In contrast to the notion that cats are either anti-social or unable to form strong social bonds with each other, they appear to be highly variable in their preferences for social interactions. Cats have been shown to have preferred associates (Crowell-Davis et al., 2004) based on observations that some individuals living in groups spend more time with one another than would be expected by chance. Preferred associates demonstrate behaviors such as allorubbing and grooming, touching while sleeping, nose touching, and signaling with a 'tail up' posture when approaching one another.

Many naturally occurring cat colonies are structured around related females. These cats show more friendly behaviors with one another than cats who are unrelated, spending more time within 1 m of conspecifics that were related than those with whom they were unfamiliar or unrelated (Curtis et al., 2003). Likewise, among housecats being temporarily sheltered in a cattery, related pairs have been found to show more affiliative behaviors than pairs of unrelated cats (Bradshaw and Hall, 1999). Unrelated cats who have lived together, and are therefore familiar with each other, are less likely to display aggressive behaviors toward each other than unrelated, unfamiliar cats, suggesting that not only relatedness, but lifetime experience and familiarity are also important predictors of successful social interactions (Barry and Crowell-Davis, 1999).

While research has demonstrated that cats can and often do form affiliative social relationships with each other, the potential for social conflict which may manifest as threat or fighting behavior between them is still of significant concern, as it is cited as one of the most common behavioral reasons that cats are rehomed or surrendered to shelters (Patronek et al., 1996; Houpt, 1998; Salman et al., 2000; Casey et al., 2009; Heath, 2009; Clark, 2016). A retrospective, survey-based study of individuals who had recently adopted a new cat reported that up to 50% of introductions between the new cat and a resident cat involved some incidents of biting and scratching. The conflict between cats was not

resolved in 35% of homes at the time of the survey (Levine et al., 2005). Another survey of pet owners with up to four cats found that over 44% of respondents reported daily signs of conflict between their cats (Elzerman et al., 2020).

Observation of such interactions may in part explain many people's misperceptions of cats as not being 'social' or having aversions to social interactions with each other. In cases of extreme conflict, living in a multi-cat household may be an underdiagnosed source of distress for some cats, ultimately undermining their overall welfare. However, even without factoring in social conflict, the effect of number of cats in a household on their physiological indicators of distress is unclear. Lichtsteiner and Turner (2008) reported that the basal urinary cortisol levels of cats in homes were not influenced by number of cats in the household. Similarly, Ramos et al. (2013) found no differences between fecal glucocorticoid levels in cats living in a multi-cat household and those housed alone, except for younger cats living in multi-cat households, who had lower fecal glucocorticoid concentrations. Additionally, in a systematic review of studies examining the relationships between cat group sizes and welfare outcomes in homes, Finka and Foreman-Worsley (2021) observed many inconsistencies in the differences in their respective stress hormone levels. They noted that factors such as handling and husbandry differences as well as disruptions in the cats' environments may have had as much or more effect on the cats' physiology as the number of cats in the household. Furthermore, they found several inconsistencies in measures and methodologies across studies, and results may have been impacted by a self-selection bias in participants. It is therefore difficult to draw conclusions about whether individual cats should live alone or with other cats. What is clear is that, collectively, the scientific evidence suggests that inter-cat sociality should be viewed as a spectrum that is influenced by several factors including relatedness, adequate access to resources (including food and space), and previous experience.

These observations fit well with what is known about facultatively social species. Consequently, decision-making about cat social living should incorporate understanding of the specific individual(s) in question based on available history, behavioral observations, and relatedness to minimize behavioral and physiological indicators of distress and maximize potential for harmonious social interactions and living conditions.

Myth 2: Cats are not social with people and therefore require minimal human social interactions

Despite their popularity, cats are still often thought of as less social, less capable of feeling emotion, and less trainable than dogs. For example, one study reported that pet owners in Japan viewed both dogs and cats as members of their family, but cats were rated lower than dogs in terms of 'intelligence' and 'ability to read humans' affective status' (Arañori et al., 2017). Owners were less likely to believe that cats experienced sadness, friendship, sympathy, compassion, or pity compared to dogs, but believed cats were equally likely to experience more negative emotions such as 'jealousy' and 'hatred' (Arañori et al., 2017).

As is the case of inter-cat sociality, the social behavior of pet cats towards humans can also vary. However, an increasing number of studies demonstrate that cats not only have the capacity to form strong social bonds with humans (e.g., Vitale et al., 2019 showed that 68% of cats studied were securely attached to their owners), they also show great social sensitivity; e.g., they preferentially approach attentive humans and follow human points or gazes to a location containing concealed food (Miklósi et al., 2005; Saito et al., 2019; Vitale and Udell 2019; Pongrácz et al., 2019). Furthermore, Vitale Shreve et al. (2017) found that cats often exhibit preferences for social stimuli when given the choice. In this study, cats were given a preference assessment where they were able to freely access options among multiple categories of choices. After identifying their favorite food, scent, toy, and human

interaction style independently, cats were given the opportunity to choose between their favorite options within each of these categories. On average cats spent about 65% of trial time in proximity to their owners during this phase. At the individual level, 50% of cats chose to spend time engaged with their human owners preferentially compared with available food, toys, and scent items during this period (Vitale Shreve et al., 2017). Further, pet cats form attachment bonds to human caretakers, forming secure attachments at roughly the same rate as has been observed in dog-human bonds and human infant-caregiver bonds (Vitale et al., 2019). Cats have also been shown to recognize when a human says their name (Saito et al., 2019), engage in social referencing (Merola et al., 2015), use some forms of synchronized non-verbal communication with human owners, such as slow blink-matching, wherein a cat opens and closes its eyes repeatedly in a series of slow blinks more frequently when their owners blink at them than when no such social interaction is taking place (Humphrey et al., 2020).

While there has been much less research on the social behavior and cognition of cats than on dogs, the research to date suggests that cats may have a great capacity for social bonding and responsiveness (Turner, 2017; Vitale Shreve et al., 2017; Vitale et al., 2019). Separation anxiety is much less likely to be diagnosed in cats than in dogs, even though there is evidence to suggest that cats can suffer from this condition (Schwartz, 2002). In attachment tests, cats show almost identical search patterns, vocalizations, and other signs of separation distress to dogs upon an owner's departure (Vitale et al., 2019).

To illustrate how human actions may influence the social behavior of cats, Vitale and Udell (2019) compared how much time cats sought proximity to an attentive human (calling the cat by name, keeping the cat's attention, and attempting contact during the full testing period) vs. an inattentive human (instructed to remain quiet, ignore the cat, and look at the floor). Shelter and pet cats spent less than 20% of the 2-min testing period interacting with an inattentive human, but on average spent over 60% of trial time seeking the proximity of the same human when the human was attentive. Collectively, these findings indicate the importance of the cat's lifelong social exposures on their social behaviors, which are more plastic than may be believed. In turn, cats' social behaviors may influence people's perceptions of their corresponding behavioral needs, creating a feedback loop that may significantly affect their welfare outcomes.

Myth 3: Socialization opportunities are less important for cats than they are for dogs

Socialization is a process of positive exposure to various stimuli, including other animals (both conspecifics and other species, including humans), sounds, experiences (e.g., wearing a harness and lead or visiting a veterinary clinic) that help a domestic pet be more prepared for life in a home with humans and/or other animals. While there is widespread availability of puppy socialization classes, dog parks, 'doggie daycares', and 'happy visit' appointments to veterinary clinics that emphasize a dog's need for early, positive, and broad social experiences, similar socialization opportunities for cats are comparatively rare. Cats are often provided few opportunities to develop their social skills, especially when caretakers assume that they do not need exposure to novel stimuli or the opportunity to experience diverse social interactions. In contrast, scientific evidence suggests that early exposure to other cats, humans, and a variety of environmental stimuli can benefit a cat's social behavior and overall cognitive development (Vitale Shreve and Udell, 2015). For example, cats with increased social exposure to humans not only display more proximity-seeking and social

responsiveness (Vitale et al., 2019), but they also demonstrate superior problem-solving abilities⁵.

The initial sensitive period for socialization in cats ranges from 2 to 7 weeks of age (Karsh and Turner, 1988). During this time, many cats would likely have exposure to their mothers and littermates, and if kept as pets, they would also have experience interacting with people. However, research in other species suggests that this limited exposure may be insufficient for cats to achieve their full social potential. For example, early research conducted with dogs found a similar sensitive period for social exposure to humans, peaking around 7 to 8 weeks of age (Scott and Fuller, 1965). Socialization through about 16 weeks (Coppinger and Coppinger, 2001) has since been identified as critical to the level of social development observed in dogs (Udell, Dorey and Wynne, 2010). Dogs with minimal exposure to conspecifics or humans during this period are typically more fearful and cautious of humans when compared to dogs with more socialization (Scott and Fuller, 1965). Similarly, research has shown that kittens who experience more exposure (e.g., 40 vs. 15 min of handling a day) to multiple humans early in life are more social and less fearful than those without this added experience (Collard, 1967; Karsh and Turner, 1988; Lowe and Bradshaw, 2002).

It is also important to note that there is incredible variability in how cats are raised and kept, which can lead to very different socialization histories. Some cats may be kept indoors and only leave their homes for veterinary exams, preventing exposure to novel social and environmental stimuli except during stressful events. This early lack of exposure may increase the level of negative arousal cats experience when new sights, sounds, scents, and other stimuli are introduced to them later in life.

Many house cats are not leash trained or actively exposed to animals outside the household at an early age, even though these beneficial opportunities can be provided safely and effectively to them (Vitale et al., 2019). At the other extreme, some pet cats are allowed to roam freely outside from a young age and have a broad exposure to many animals and stimuli. Many cats' experiences fall somewhere in between. Opportunities for socialization (to other cats, other animals, humans, and to environmental stimuli) vary greatly among cats, and this may contribute to the significant variability in sociability observed within the species (Vitale Shreve and Udell, 2015). Further research is required to ascertain the relationship between cats' socialization histories and their social development. However, given burgeoning evidence that a cat's history of social exposure and access to diverse social stimuli early in life is a significant predictor of their social behavior and welfare, it is imperative to highlight the importance of socialization as a potential tool to improve cat welfare (Vitale et al., 2019).

Welfare implications

The 'myth-conceptions' of the domestic cat's capacity for social behavior and corresponding needs hold profound implications for cat care and welfare. Maintaining cats in environments that inadequately address their early and ongoing socialization needs, or that fail to consider sources of social conflict, and strategies for avoiding or mitigating it, may contribute to chronic social stress, and ultimately, poor health and welfare (Stella et al., 2013; Stella and Buffington, 2014; Stella and Croney, 2016).

Cats also appear to have diverse coping styles (Stella and Croney, 2019) which may manifest differently depending on the stressors they encounter. Coping styles have been defined by Koolhaas et al. (1999) as 'a coherent set of behavioral and physiological stress responses which is

consistent over time.' In a study of 55 owned cats, Stella and Croney (2019) identified two clusters of cats that appeared to cope differently with the stressor of short-term cage confinement in a laboratory. The first, described as 'shy, calm, mellow, and timid' by their owners ate more, hid more and had longer latencies to approach familiar and unfamiliar people compared to their counterparts. The second cluster (n = 33), characterized as 'active, playful, curious, and easygoing' showed more affiliative behaviors toward experimenters, and they vocalized, and perched more frequently on boxes offered as enrichment.

Cats may experience distress when their individuality (including their personal history of lifetime experiences) is not appropriately considered. This may be especially true when a cat is rehomed, or when new people or animals, including other cats, appear in their established household or territory. It can also be difficult to gauge how cats may cope with distress resulting from being kept in living situations where they cannot easily escape undesired social contact with each other. One reported challenge is that the signals cats use for diffusing conflict are less well understood than those of dogs and they may also be less visually obvious to owners (Rochlitz, 2009). This may make resolution of conflicts more challenging once they begin. In the home, a cat's capacity to cope with their social environment may be challenged when they must also contend with life in close proximity to neighborhood cats (Bradshaw, 2016).

False beliefs that cats are asocial by nature, may result in owners expecting or accepting social fear, anxiety, or avoidance in their cats despite the negative welfare consequences of these states. This can lead to missed opportunities to support better social outcomes for cats, or a failure to look for solutions. Fortunately, empirically guided solutions to some inter-cat conflicts exist. For example, increasing access to areas where cats can avoid one another when needed, and providing adequate resources (food, water, litterboxes, preferred sleeping locations) to support the number of cats sharing a living space can reduce the potential for conflict (Bernstein and Strack, 1996; Heidenberger, 1997). Many owners, however, are not aware of these needs, and consequently fail to provide safe spaces, the recommended number of litter boxes, or enough feeding stations and other resources necessary to reduce competition and conflict (Heidenberger, 1997). This may not only strain inter-cat relationships and increase distress, but may also promote disease processes, such as idiopathic cystitis within multi-cat households (Kim et al., 2018).

Bradshaw (2016) addresses another frequently encountered myth embedded in many owners' misunderstanding of cat social behavior - namely that all singly kept cats are lonely and could benefit from having another cat brought into the home. While in some instances, cats may indeed benefit from company of their own kind, domestic cats have diverse social compositions that include both solitary and social group structures. Because many factors such as access to resources and early socialization experience influence the likelihood of cats enjoying or tolerating each other, it can be difficult to predict how unfamiliar individuals may respond when introduced in homes. More research is needed on biological, environmental, and other factors that contribute to individual differences in cat social behavior to help address such challenges.

Cats can also experience distress when their social needs in relationships with humans are not met. For example, cats with fewer interactions with people (Grigg and Kogan, 2019) and longer periods of time left alone (Heidenberger, 1997) displayed higher numbers of problematic behaviors (e.g., inter-cat aggression) and were perceived to have higher anxiety levels. In addition, cats separated from bonded human caretakers in an unfamiliar place (such as a kennel or veterinary office) even for a short time exhibited signs of separation distress, with the quality of the bond shared between owner and cat serving as a predictor of how an individual cat might react upon their owner's return (Behnke et al., 2021).

Some studies have shown that increased human presence and social activity are associated with higher stress in cats, as indicated by their

⁵ See: Howard, M., 2018. Socialization and problem-solving in domestic cats (*Felis catus*). Masters Theses and Doctoral Dissertations. University of Tennessee. Chattanooga, TN, USA. <https://scholar.utc.edu/theses/550>. (Accessed 29 August 2023)

serum glucocorticoid concentrations (Lichtsteiner and Turner, 2008; Ramos et al., 2013). However, individual preferences for human social contact may mediate this effect, as Ramos et al. (2013) found an association between higher glucocorticoid levels and a cat being described as 'merely tolerating', rather than 'enjoying or seeking out petting,' by owners.

These findings highlight the diversity and complexity of cat social interactions with humans, indicating that many do form strong, secure bonds with people and that human-cat interactions can have important welfare implications for cats (Turner, 2017; Vitale Shreve et al., 2017; Vitale et al., 2019). Educating owners about socialization, positive social contact, and how to facilitate strong, mutually beneficial relationships with cats is essential to enhance their welfare.

Cats' socialization windows occur early in their lives. Therefore, efforts should be made to target animal shelters, foster caretakers, and new adopters of kittens to ensure that kittens are given adequate and positive exposure to a variety of stimuli. Proper socialization prepares kittens for future interactions with both humans and other cats.

Conclusions

Owner education about cat sociality and its relationship to their quality of life is essential, given that cat behavioral problems, which are often rooted in fear, anxiety, and social stress, are regularly cited as reasons for their abandonment, relinquishment, and euthanasia (Patronek et al., 1996; Salman et al., 2000; Coe et al., 2014). Veterinarians may be able to further support cat welfare by educating owners about the relationships between the cat's social environment, stress, and health.

Counseling owners to ensure they are aware of cats' social needs and have sufficient knowledge to monitor and identify related indicators of their well-being is critical to avoid negative affective states and behaviors that undermine both the human-cat bond and cat welfare outcomes. Increasing research and education efforts relating to the social behavior of cats may significantly improve the lives of cats, and the bonds we share with them.

Declaration of Competing Interest

None of the authors of this paper has a financial or personal relationship with other people or organizations that could inappropriately influence or bias the content of the paper.

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